Callan McGill

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Education

- MMath Mathematics University of Warwick, First-Class Honours 2008-2012.
- PhD Mathematics (Incomplete), Sheffield University 2012-2016.

Employment

- **Bloomberg**, *Senior Software Engineer* [March 2023 - Present]
  - Working for the static analysis team, which is reponsible for implementing company-wide policies and initiatives by creating custom linting tools that feed into our automated re-factoring infrastructure.
  - Created tooling to perform static analysis of Fortran code based on the Haskell *fortran-src* package, including automated feature flag removal and dead code elimination.
  - Created a C++ rewrite rules engine in Rust based on *treesitter* to aid in automated feature flag removal and dead code elimination.
  - Working on a Bloomberg semantic code search project using *Glean*.
  - One of two members of the Haskell infrastructure at Bloomberg.

- **Symbiont IO**, *Language Engineer* [June 2021 - November 2022]
  - Worked primarily on the frontend and elaborator of the SymPL programming language - a safe, statically-typed subset of Python with builtin cryptographic and database capabilities. The elaborator translates from this language to a core System-F-style language
  - Wrote a *rustc*-inspired error reporting engine from scratch and vastly improved the error messages and user-experience of the SymPL language.
  - Implemented a configurable primtitive effects system in the SymPL virtual machine.

- **American Museum of Natural History**, *Researcher and Programmer* [July 2018 - July 2020]
  - Designed, wrote and refined a novel phylogenetics analysis platform in Haskell. The platform can perform both tree and network optimality searches using a variety of search strategies (Wagner build, genetic algorithms, graph moves, Bayesian optimization).
  - Wrote novel distance methods and clustering algorithms for Phylogenetic Networks, thus allowing for massively parallel Wagner builds and discovering significantly faster initial solutions.
  - Organized and oversaw summer projects for interns from Columbia University and NYU.

- **Oxford Nanoimaging**, *Research Programmer* [Jan - May 2018]
  - Wrote the build system for our C++ projects in Haskell using Shake.

  - Wrote and implemented the specification for serialisation and deserialisation of our custom microscope data format in binary and JSON using C++.

- **University of Sheffield**, *Teaching Assistant* [2012 - 2017]
  - Taught undergraduate courses in Calculus, Linear Algebra and Discrete Mathematics. Assisted in grading assignments and exams, holding office hours, and offering additional help to students.
  - Volunteered as a one-to-one tutor in the Mathematics and Statistics help centre where I worked with undergraduates informed by modern pedagogical methods.

Technical Skills

- **Experience**:
  - **Writing**: Working on an in progress book on topos theory for type theorists which uses Agda as a pedagogical tool to formalize some parts of the book.
  - **Core library maintainer**: Served as one of the maintainers of the Haskell *text* library.
  - **Open source contributor**: Have made contributions to Haskell open source projects and core libraries including *vector*, *containers*, *parallel*, *aeson* , *bytestring* and *extra*. I have also contributed to the *Agda* standard library and *agda-categories*.
  - **Contributor & presenter**: I was one of the organizers of the Homotopy Type Theory NYC reading seminar: I presented on various topics in category theory, topos theory, and homotopy type theory. I am currently one of the organizers for a study group at the CUNY graduate center on the philosophy of higher order logic.
  - **Freelance Software Engineer**: Contributed to a diverse array of projects from individuals and small companies, including work in Haskell, Python, OCaml, Rust and Prolog, as well as mathematical assistance.
- **Programming Languages** :
  - **Fluent:** Haskell, Agda, OCaml, Python, Rust, C++
  - **Competent:** Emacs Lisp, C, Coq, Lean4, LaTeX
  - **Familiar:** Bash, R, Nushell
- **Courses/Conferences Attended**:
  - Conference 2022: Oregon Programming Languages Summer School
  - Conference 2020: Formal Methods for the Informal Engineer, The Z3 Theorem Prover
  - Conference 2020: Formal Methods for the Informal Engineer, The Coq Theorem Prover
  - Conference 2019: Compose, Speaker: *"Phylogeneric Software in Haskell"*
  - Conference 2016: West Coast Algebraic Topology Summer School, Speaker: *"Building Cohomology Theories From P-Divisible Groups"*
  - Course, 2021: Normalization for Type Theory by Aaron Stump
  - Course, 2020: The Implementation of Idris 2 by Edwin Brady
- **Academic Proficiencies**:
  - **Mathematics:** The Weil Conjectures, stable homotopy theory, topos theory
  - **Computer Science:** Formal verification, Compilers, Dependent type theory
  - **Music:** Studying classical guitar performance for over ten years